

MaskCRT-B: Masked Conditional Residual Transformer for Learned B-frame Coding

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Abstract—This paper proposes a learned hierarchical B-frame coding scheme in response to the Grand Challenge on Neural Network-based Video Coding at ISCAS 2025. Recently, masked conditional residual coding emerged as an attractive alternative to the existing inter-frame coding frameworks, including residual coding, conditional coding, and conditional residual coding. In this work, we propose masked conditional residual B-frame coding, termed MaskCRT-B, for YUV420 videos. It features an asymmetric codec architecture that includes one joint YUV encoder and two separate Y and UV decoders. Moreover, it incorporates a bi-directional adaptive fusion module that refines the bi-directional feature maps to better tackle the prediction of the occluded and dis-occluded regions within the input video. MaskCRT-B presents a significant advancement in learned B-frame coding, outperforming the state-of-the-art conditional B-frame codec from the Grand Challenge at ISCAS 2024.

Index Terms—Learned video coding, YUV420, B-frame coding

I. INTRODUCTION

Learned video coding [1]–[8] has made remarkable progress, with some state-of-the-art approaches [5]–[7] performing comparably to or surpassing H.266/VVC under the low-delay configuration with P-frame coding. Currently, there are four main inter-frame coding frameworks depending on how the temporal predictor x_c is utilized. These are (1) residual coding [1], [2], (2) conditional coding [3]–[6], (3) conditional residual coding [8], and (4) masked conditional residual coding [7]. Residual coding follows the traditional coding architecture to encode an input video frame x_t in the form of residual signals $x_t - x_c$. Conditional coding, which has gone mainstream, further extends the simple linear prediction of x_t to potentially a non-linear prediction by learning a conditional variational autoencoder. Although theoretically superior to residual coding, conditional coding may suffer from the information bottleneck issue [9]. This motivates conditional residual coding [8]. More recently, masked conditional residual coding [7] is proposed to switch between conditional coding and conditional residual coding at the sub-frame level, in order to achieve the best of both worlds.

While the aforementioned frameworks are developed primarily for P-frame coding, several attempts have been made to develop their B-frame extensions. B-frame coding differs from P-frame coding in that it is allowed to leverage both past and future reference frames for coding. Pourreza *et al.* [10] adopt a straightforward extension by reusing the P-frame codec for B-frame coding. They first interpolate a temporal predictor

from the two reference frames, followed by using it as an ordinary reference frame for coding an input frame in the same manner as coding a P-frame. The works in [11]–[13] follow the residual coding framework to signal optical flow maps and residual signals in the bitstream. Particularly, Yilmaz *et al.* [11] and Çetin *et al.* [14] apply residual coding to both motion and inter-frame coding. In comparison, Wu *et al.* [15] present an early attempt at conditional B-frame coding by leveraging the multi-scale features from the reference frames as the condition. Chen *et al.* [3], [16] and Gao *et al.* [17] additionally signal motion information and adopt a GridNet-like structure [18] to fuse the motion-compensated, multi-scale features. For better motion compensation, Yilmaz *et al.* [19], [20] employ deformable convolutions. To save the motion overhead, Alexandre *et al.* [21] adopts a two-layer CANF [22] without motion coding. More recently, Sheng *et al.* [23] and Yang *et al.* [24] extend straightforwardly temporal context mining [5], [6] to B-frame coding by learning temporal context from both directions. Despite their promising results, the implicit buffering schemes incur a large buffer size and high memory access bandwidth.

Motivated by the study in [25], where masked conditional residual coding stands out from the other mainstream inter-frame coding frameworks in terms of the coding efficiency, buffer size, model size, and computational complexity, we propose a novel B-frame coding framework (termed MaskCRT-B) based on masked conditional residual coding. We draw inspiration from MaskCRT [7], which is designed for P-frame coding and the RGB content. To the best of our knowledge, this work is the first attempt at developing learned B-frame YUV coding with the masked conditional residual coding framework. Our MaskCRT-B has the following features. (1) It has an asymmetric codec architecture that includes one joint YUV encoder and two separate Y and UV decoders. (2) It incorporates a bi-directional adaptive fusion module that aims for refining the bi-directional feature maps to better tackle the prediction of the occluded and dis-occluded regions within the input video. Experimental results show that MaskCRT-B presents a significant advancement in learned B-frame coding, outperforming the state-of-the-art conditional B-frame codec [17] submitted to the Grand Challenge at ISCAS 2024.

II. PROPOSED METHOD: MASKCRT-B

A. System Overview

Fig. 1 (a) depicts our MaskCRT-B framework. As shown, encoding a B-frame x_t^{420} in the YUV420 format, where

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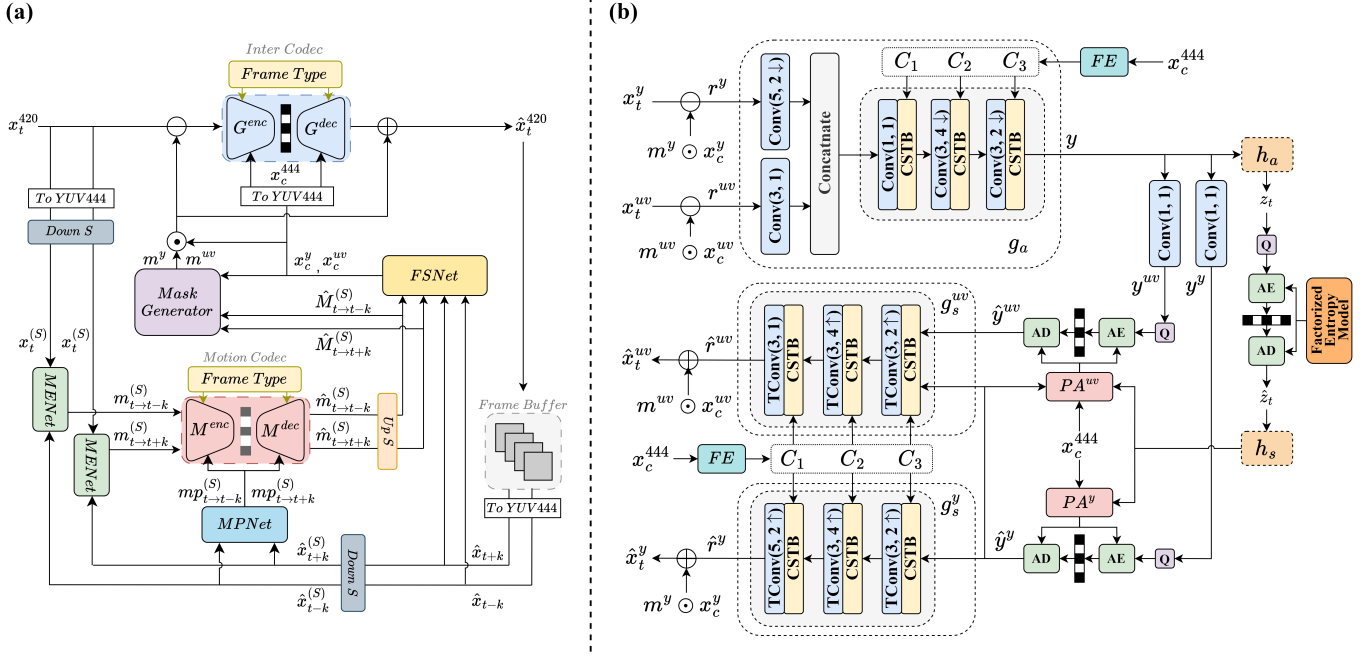


Fig. 1: (a) Illustration of our MaskCRT-B framework. The operation "To YUV444" converts an input YUV420 frame into its YUV444 format. *Down S* and *Up S* denote the bilinearly downsampling and upsampling with the downsampling factor S , respectively. (b) The network architecture of our inter-frame codec, where *FE*, *PA*, and *CSTB* are the feature extractor, the prior aggregation module, and the conditional Swin-Transformer blocks [7], respectively.

the Y component is of height H and width W , starts with performing motion estimation based on an off-the-shelf motion estimation network (MENet). The pre-trained MENet is adapted to the YUV444 content with a minimal change to its network architecture. Notably, to mitigate the domain shift issue, where the MENet trained on small group-of-pictures (GOPs) may struggle to handle large motion in large GOPs during inference, we follow OMRA [26] to adapt the motion resolution by choosing a downsampling factor S from $\{1, 2, 4, 8\}$ for motion estimation and coding. The S is determined through online rate-distortion optimization. More specifically, the target frame and reference frames are downsampled by a factor of S to obtain low-resolution frames $x_t^{(S)}, \hat{x}_{t-k}^{(S)}, \hat{x}_{t+k}^{(S)} \in \mathbb{R}^{3 \times H/S \times W/S}$, before generating two low-resolution optical flow maps, $m_{t \rightarrow t-k}^{(S)}, m_{t \rightarrow t+k}^{(S)} \in \mathbb{R}^{2 \times H/S \times W/S}$. These flow maps are compressed by the motion codec $\{M^{enc}, M^{dec}\}$, where the motion predictors $mp_{t \rightarrow t-k}^{(S)}, mp_{t \rightarrow t+k}^{(S)} \in \mathbb{R}^{2 \times H/S \times W/S}$ are output by the motion prediction network (MPNet), which takes also the downsampled reference frames $\hat{x}_{t-k}^{(S)}, \hat{x}_{t+k}^{(S)} \in \mathbb{R}^{3 \times H/S \times W/S}$ as inputs. The reconstructed flow maps $\hat{m}_{t \rightarrow t-k}, \hat{m}_{t \rightarrow t+k} \in \mathbb{R}^{2 \times H/S \times W/S}$ are bilinearly upsampled to match the resolution of the Y component in x_t^{420} , allowing motion compensation to be performed at the highest resolution. These upsampled flow maps $\hat{M}_{t \rightarrow t-k}, \hat{M}_{t \rightarrow t+k} \in \mathbb{R}^{2 \times H \times W}$, along with $\hat{x}_{t-k}, \hat{x}_{t+k}$, are fed to the frame synthesis network (FSNet), producing the temporal predictors $x_c^y \in \mathbb{R}^{1 \times H \times W}$, $x_c^{uv} \in \mathbb{R}^{2 \times H/2 \times W/2}$ in the YUV420 format. The x_c^y , x_c^{uv} and $\hat{M}_{t \rightarrow t-k}, \hat{M}_{t \rightarrow t+k}$ are used in the mask generator to arrive at two independent pixel-wise soft masks $m_t^y \in \mathbb{R}^{1 \times H \times W}$, $m_t^{uv} \in \mathbb{R}^{2 \times H/2 \times W/2}$

for the Y and UV components, respectively. These soft masks allow our codec to switch between conditional coding and conditional residual coding at the pixel level. The inputs to the inter-frame codec $\{G^{enc}, G^{dec}\}$ are $(1 - m_{\{y, uv\}}) \odot x_t^{\{y, uv\}} + m_{\{y, uv\}} \odot (x_t^{\{y, uv\}} - x_c^{\{y, uv\}})$, where $\odot$ denotes the element-wise multiplication. For simplicity, the condition signals for the inter-frame codec are in the YUV444 domain to match the resolution of the internal features. These signals are derived from the temporal predictor $x_c^{444} \in \mathbb{R}^{3 \times H \times W}$ converted from x_c^y, x_c^{uv} . Finally, the reconstructed frame $\hat{x}_t^{420}$ is obtained by adding $m_{\{y, uv\}} \odot x_c^{\{y, uv\}}$ to the output of their respective inter-frame decoders. For better bit-allocation, both the motion and inter-frame codecs adopt a frame-type condition signal [3], [16], [17] to distinguish between reference and non-reference B-frames.

B. Inter-frame Codec for YUV420 Content

Fig. 1 (b) details our masked conditional residual inter-frame codec for coding the YUV420 content. It features one joint encoder g_a to make full use of the correlation among these color components for encoding, and two separate decoders g_s^y, g_s^{uv} to optimize individually the reconstruction of the Y and UV components. For masked conditional residual coding, the input residual signals $r^y \in \mathbb{R}^{1 \times H \times W}$, $r^{uv} \in \mathbb{R}^{2 \times H/2 \times W/2}$ are given by

$$r^y = x_t^y - m_t^y \odot x_c^y, \quad (1)$$

$$r^{uv} = x_t^{uv} - m_t^{uv} \odot x_c^{uv}, \quad (2)$$

where $\{x_t^y, x_t^{uv}\}$, $\{m_t^y, m_t^{uv}\}$, and $\{x_c^y, x_c^{uv}\}$ are the target frames, masks, and temporal predictors of the Y and UV

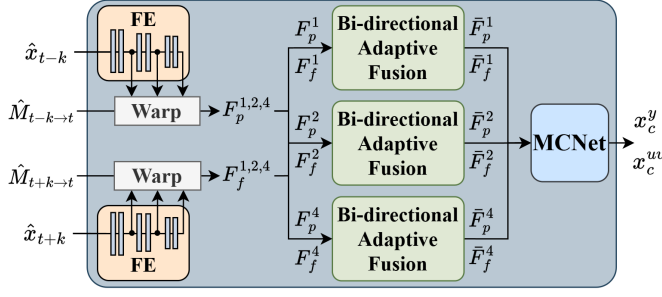


Fig. 2: Illustration of our frame synthesis network. FE represents the feature extractor used to generate multi-scale features $F_p^{1,2,4}$ and $F_f^{1,2,4}$ from $\hat{x}_{t-k}$ and $\hat{x}_{t+k}$, respectively.

components, respectively. For the joint encoding of the Y and UV components, we downsample r^y via a convolutional layer to match the spatial resolution of r^{uv} . The condition signals $C_1 \in \mathbb{R}^{96 \times H/2 \times W/2}$, $C_2 \in \mathbb{R}^{128 \times H/8 \times W/8}$, $C_3 \in \mathbb{R}^{192 \times H/16 \times W/16}$ generated by the feature extractor FE based on the temporal predictor x_c^{444} in the YUV444 format are delivered into the encoder g_a and decoders g_s^y , g_s^{uv} . Our compression backbone adopts conditional Swin-Transformer blocks (CSTB) [7] to obtain the main latents $y \in \mathbb{R}^{192 \times H/16 \times W/16}$, which is subsequently split into $y^y \in \mathbb{R}^{160 \times H/16 \times W/16}$ and $y^{uv} \in \mathbb{R}^{32 \times H/16 \times W/16}$.

The distributions of both latents are predicted by prior aggregation module PA , which leverages the hyperprior and the temporal prior x_c^{444} . Particularly, for decoding, the Y latents $\hat{y}^y$ are first reconstructed and used as the additional side information to entropy decode the UV latents $\hat{y}^{uv}$. Moreover, $\hat{y}^y$ is fed into the UV decoder to jointly reconstruct the UV residues.

C. Frame Synthesis Network (FSNet)

The FSNet (see Fig. 2) fuses information from the past and future reference frames to generate the temporal predictors x_c^y , x_c^{uv} . It begins with updating the temporally warped, bi-directional frame features $F_p^{1,2,4}$, $F_f^{1,2,4}$, followed by fusing the updated features $\bar{F}_p^{1,2,4}$, $\bar{F}_f^{1,2,4}$ with a MCNet that has a GridNet-like structure [18].

The feature update aims to refine the bi-directional features by having them complement each other. We observe that in coding a B-frame by forming predictions from its past and future references, the dis-occluded regions can be better predicted from the future frame and the occluded regions from the past frame. We thus update $F_p^{1,2,4}$ with $F_f^{1,2,4}$, and vice versa. To this end, we introduce a multi-scale, gating mechanism that proceeds as follows:

$$\bar{F}_p^i = \alpha^{(i)} \odot F_p^i + (1 - \alpha^{(i)}) \odot F_f^i, \quad (3)$$

$$\bar{F}_f^i = \beta^{(i)} \odot F_f^i + (1 - \beta^{(i)}) \odot F_p^i, \quad (4)$$

where $\alpha^{(i)}, \beta^{(i)} \in \mathbb{R}^{1 \times H/i \times W/i}$, $i \in \{1, 2, 4\}$ represent the spatial-wise gating signals at scale i , and $\odot$ represents the element-wise multiplication. Notably, these spatial-wise masks $\alpha^{(i)}, \beta^{(i)}$ have real-valued elements that range between 0 and

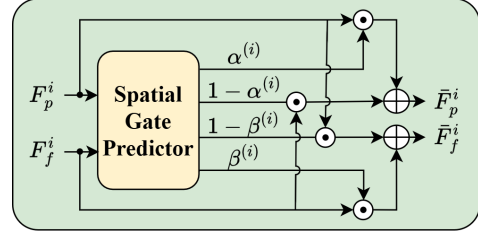


Fig. 3: The bi-directional adaptive fusion module (BiAF).

TABLE I: Comparison of the BD-rates and kMAC/pixel with and without BiAF. The number in parentheses indicates the percentage increase in kMAC/pixel with BiAF.

		HEVC-B	UVG	ISCAS'24	kMAC/pixel
w/o	BiAF	0.0	0.0	0.0	2452.80
w/	BiAF	-4.4	-5.9	-2.9	2576.86 (+5.1%)

1, and are generated by the spatial gate predictor (see Fig. 3) taking $F_p^{1,2,4}$, $F_f^{1,2,4}$ as inputs to identify the dis-occluded regions in the past and future reference frames, respectively.

III. EXPERIMENTS

We train our model on the Vimeo-90k dataset [27]. For all the 7-frame, RGB training sequences at resolution 448×256 , we convert them from RGB to YUV420 using the BT.709, and the frames are randomly cropped to 256×256 . We use Adam optimizer [28] with a learning rate of $1e^{-4}$. Our training objective and strategy are the same as [17].

We evaluate our model on UVG [29], HEVC Class B [30], and ISCAS'24 [31] test datasets. All the test sequences are in the YUV420 format and encoded with 96 frames, with the intra-period and GOP size set to 32. Following the evaluation protocol of the Grand Challenge on Neural Network-based Video Coding ISCAS 2025 [32], we report BD-rate saving in terms of PSNR-YUV, with the bit rate measured in bits per pixel (bpp). Negative and positive BD-rate numbers suggest rate reductions and inflations, respectively. The anchor is HM-16.22 [33] with the configuration *encoder_randomaccess_main.cfg*.

A. Experimental Results

1) *Bi-directional Adaptive Fusion (BiAF)*: Table I shows that our bi-directional adaptive fusion module (Fig. 3) is effective on all the test datasets, with only a modest complexity overhead. Fig. 5 (the first row) sheds light on its inner workings by visualizing the gating signals $\alpha^{(1)}$, $\beta^{(1)}$ for *Jockey* sequence. $\alpha^{(1)}$, $\beta^{(1)}$ reveal the spatial adaptive weighting for F_p^1 , F_f^1 , respectively (see Fig. 2). We note that $\alpha^{(1)}$ weights more heavily the past reference features F_p^1 on the left boundary of the racing horse, where the occlusion occurs. In contrast, $\beta^{(1)}$ assigns higher weighting to the future reference features F_f^1 on the right boundary, where the dis-occlusion occurs. The results are further corroborated by the prediction residues (in the pixel domain) that result from the single use of the past or future reference for temporal prediction (see the second row of Fig. 5).

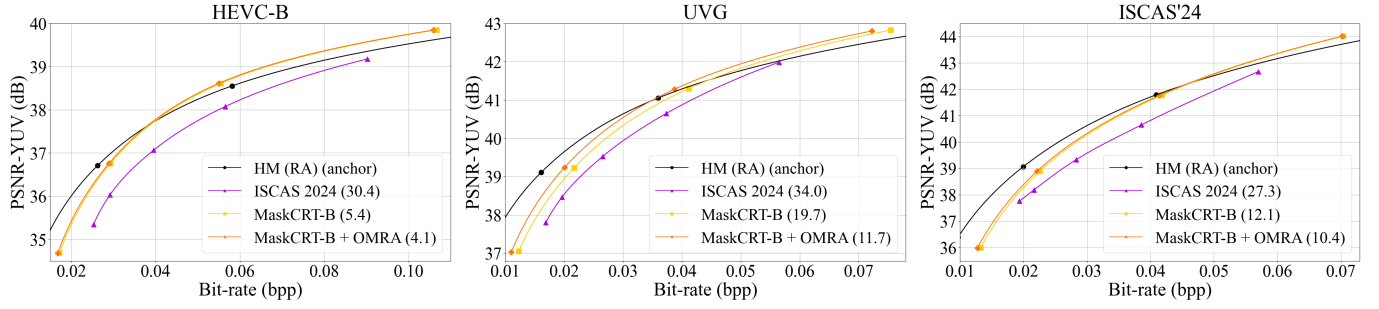


Fig. 4: Rate-distortion performance comparison with intra-period 32. The numbers within the parentheses are BD-rates, with HM-16.25 (Random Access) serving as the anchor. ISCAS 2024 refers to the learned B-frame codec [17] from the Grand Challenge at ISCAS 2024.

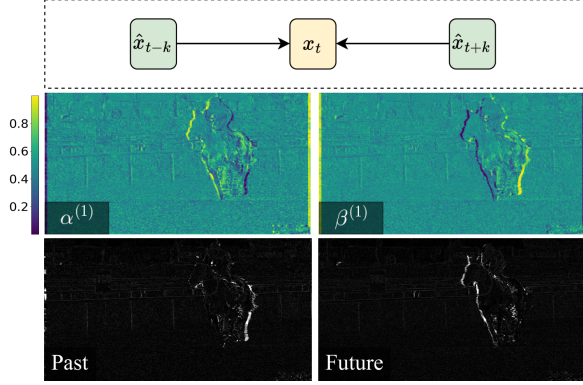


Fig. 5: Visualization for *Jockey* sequence. $\alpha^{(1)}$, $\beta^{(1)}$ indicate the reliability of the prediction from the past and future references, respectively. The second row displays the prediction residues between the coding frame x_t and the motion-compensated reference frame from the past or future.

TABLE II: Comparison of the BD-rates and kMACs/pixel: joint vs. separate decoders.

	HEVC-B	UVG	ISCAS'24	kMAC/pixel
Joint Decoder	0.0	0.0	0.0	2273.46
Sep. Decoders	-4.3	-3.9	-4.2	2576.86

2) *Rate-distortion Comparison*: Fig. 4 presents the rate-distortion comparison. We observe that (1) our MaskCRT-B significantly outperforms the learned B-frame codec from the Grand Challenge at ISCAS 2024 [17] across various datasets. (2) Further improvements are achieved by incorporating OMRA [26] into our YUV codec, with more pronounced effects in the UVG dataset, which has quite a few fast-motion sequences. (3) Although inferior to HM-16.22 [33] at lower bitrates, our MaskCRT-B outperforms HM-16.22 [33] at higher bitrates (above 0.06 bpp).

3) *Joint Decoder vs. Separate Decoders*: Table II presents the BD-rate comparison between the joint and separate YUV decoders. Both designs share the same encoder structure, but the joint variant uses two decoding heads within the single decoder to reconstruct the Y and UV components. Despite its moderately higher computational complexity, our separate Y and UV decoders consistently outperform the joint decoder, achieving 3.9%-4.3% BD-rate savings.

TABLE III: Comparison with the state-of-the-art learned video codec in terms of the BD-rate, encoding/decoding time and memory bandwidth. Memory bandwidth is measured in the number of full-resolution (the same spatial resolution as the Y component of the input frame) feature maps that must be accessed to decode a video frame. For example, retrieving a YUV420 reference frame is equivalent to 1.5 full-resolution feature maps. Enc./Dec. represents encoding and decoding time per frame, respectively.

Methods	BD-rate (%) PSNR-YUV						Enc./Dec. Time (s)	Memory Bandwidth
	HEVC-B		UVG		ISCAS'24			
	I32	I16	I32	I16	I32	I16		
MaskCRT-B	0.0	0.0	0.0	0.0	0.0	0.0	6.54/0.99	3
DCVC-FM [6]	-19.9	4.7	-29.7	-8.2	-31.4	-10.1	0.60/0.48	55.75

B. Comparison with the State-of-the-art Codecs

Table III compares MaskCRT-B with DCVC-FM [6]. For a fair comparison, both codecs use the same intra codec from [6], and are tested with intra-periods 32 and 16. As shown, DCVC-FM generally performs better than MaskCRT-B. The performance gap is seen to be smaller with intra-period 16, where the domain shift issue [26] is less severe for B-frame coding. However, it is worth noting that to decode a video frame, MaskCRT-B uses only two bi-directional reference frames in the YUV420 format, whereas DCVC-FM needs to fetch 55.75 high-resolution feature maps. This implies that DCVC-FM incurs much higher memory access bandwidth. In contrast, MaskCRT-B exhibits slightly higher encoding time due to the encoder-side optimization [26].

IV. CONCLUSION

This work presents a masked conditional residual B-frame coding framework (MaskCRT-B) for YUV420 content. It features a joint encoder, and separate Y and UV decoders. Additionally, it has a bi-directional adaptive fusion module (BiAF) that adaptively fuses the bi-directional feature maps extracted from past and future frames. Experimental results show that MaskCRT-B significantly outperforms the learned B-frame codec from ISCAS 2024 GC [17] across various datasets, and the separate Y and UV decoders prove to be superior to the joint YUV decoder. As compared with DCVC-FM, the state-of-the-art learned P-frame codec, MaskCRT-B has much smaller memory access bandwidth.

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